37 Metal-matrix composites

Metal-matrix composites are engineered materials typically comprising reinforceants of high elastic modulus and high strength in a matrix of a more ductile and tougher metal of lower elastic modulus and strength. The metal-matrix composite has a better combination of properties than can be achieved by either component material by itself. Normally, the objective of adding the reinforceant is to transfer the load from the matrix to the reinforceant so that the strength and elastic modulus of the composite are increased in proportion to the strength, modulus and volume fraction of the added material. Improvements in non-mechanical properties can also be achieved by adding a second phase, for example electrical conductivity, as compared with conventional alloying for strength.

The reinforcement can take one of several forms. The least expensive and most readily available on the market are the particulates. These can be round but are usually irregular particles of ceramics, of which SiC and Al_2O_3 are most frequently used. Composites reinforced by particulates are isotropic in properties but do not make best use of the reinforceant. Fine fibres are much more effective though usually more costly to use. Most effective in load transfer are long parallel continuous fibres. Somewhat less effective are short parallel fibres. Long fibres give high axial strength and stiffness, low coefficients of thermal expansion and, in appropriate matrices, high creep strength. These properties are very anisotropic and the composites can be weak and brittle in directions normal to that of the fibres. Where high two-dimensional properties are needed, cross-ply or interwoven fibres can be used. Short or long randomly oriented fibres provide lower efficiencies in strengthening (but are still more effective than particulates). These are most frequently available as SiC whiskers or as short random alumina ('Saffil') fibres or alumino-silicate matts.

Long continuous fibres include drawn metallic wires, mono-filaments deposited by CVD or multifilaments made by pyrolysis of polymers. The properties of some typical fibres are compared in Table 37.1. The relative prices are given as a very approximate guide. Because most composites are engineered materials, the matrix and the reinforceant are not in thermodynamic equilibrium and so at a high enough temperature, reaction will occur between them which can degrade the properties of the fibre in particular and reduce strength and more especially fatigue resistance. As many composites are manufactured by infiltration of the liquid metal matrix into the pack of fibres, reaction may occur at this stage. Some typical examples of interaction are listed in Table 37.2.

In order to obtain load transfer in service, it is essential to ensure that the reinforceant is fully wetted by the matrix during manufacture. In many cases, this requires that the fibre is coated with a thin interlayer which is compatible with both fibre and matrix. Commonly, this also has the advantage of preventing deleterious inter-diffusion between the two component materials. The data on most coatings are proprietary knowledge. However, it is well known that silicon carbide is used as an interlayer on boron and on carbon fibres to aid wetting by aluminium alloys.

The routes for manufacturing composites are still being developed but the most successful and lowest cost so far is by mixing particulates in molten metal and casting to either foundry ingot or as billets for extrusion or rolling. This is applied commercially to aluminium alloy composites. Another practicable route is co-spraying in which SiC particles are injected into an atomised stream of aluminium alloy and both are collected on a substrate as a co-deposited billet which can then be processed conventionally. This is a development of the Osprey process and can be applied more widely to aluminium and other alloys. Other routes involve the infiltration of molten metal into fibre pre-forms of the required shape often contained within a mould to ensure the correct final shape. This can be done by squeeze casting or by infiltrating semi-solid alloys to minimise interaction between the fibre and metal. Fibres can also be drawn through a melt to coat them and then be consolidated by hot-pressing.

To reduce interaction, solid state methods can be used, e.g. cold isostatic pressing (CIP) and sintering or hot isostatic pressing (HIP) of metal powders mixed with short fibres. Diffusion-bonding

Table 37.1 PROPERTIES OF REINFORCING FIBRES AT ROOM TEMPERATURE (BASED ON REFERENCES 9 and 16)

Fibre	*Form*	*Preparation route*	*Diameter* μm	*Density* g cm^{-3}	*Weibull modulus*	*Fracture stress* MPa	*Elastic modulus* GPa	*Coefficient of thermal expansion* $K^{-1} \times 10^6$	*Price relative to glass fibre*
Tungsten	Wire	Drawn	10–500	19.2		2 500	400	5.0	1 000–50
Steel	Wire	Drawn	10–250	7.8		2 500	210	15.0	30–1
Boron	Cont. mono-filament	Chemical vapour depos.	150	2.6	7–19	3 500	400	8.0	250
SiC	Cont. mono-filament	Chemical vapour depos.	150	3.4	8	3 800	450	4.5	500
SiC	Cont. multi-filament	Polymer fibre pyrolysis	10–15	2.6		2 500	200	4.5	100
SiC	Whisker (random, short)	Polymer fibre pyrolysis	0.1–2.0	3.2	1.8	10 000	700	4.5	150
~Al_2O_3	Multi-filament	Oxide/salt fibre	15–25	3.9		1 500	380	7.0	100
~Al_2O_3	Random short fibres	pyrolysis	2–4	3.5	6	2 000	300	7.0	25
C (high modulus)	Cont. multi-filament	Polymer fibre pyrolysis	10	2.0	3–8	3 000	600	0	1 000
C (med. strength)	Cont. multi-filament	Polymer fibre pyrolysis	8	1.9	5–8	4 200	300	0	100
Alumino-silicate Glass (27% SiO_2)	Random short fibres	Polymer fibre pyrolysis	3	3.0		850	150	—	20
Alumino-silicate Glass (47% SiO_2)	Random short fibres	Drawn from melt	3	2.7		1 750	105	—	1
S-Glass	Cont. multi-filament	Drawn from melt	3–20	2.5		4 000	90	3.0	2.5

Table 37.2 TYPICAL INTERACTIONS IN SOME FIBRE-MATRIX SYSTEMS

System	*Potential interaction*	*Temperature of significant interaction* °C	*References*
Al-C	Formation of Al_4C_3 at interface. Degradation of C fibre properties.	550 ~495	9
Al-Al_2O_3	No significant reaction at normal fabrication temperatures		9
Al-oxide (Al_2O_3-SiO_3-B_2O_3)	B_2O_3 reacts with Al to form borides.	770	10
Al-B	Boride formation; interlayer of SiC needed.	500	9
Al/Li-Al_2O_3	Interfacial layer of $LiAl_5O_8$ on liquid infiltration.	~650	13
Al-SiC	No significant reaction below melting point. Al_3C_3 and Si can form in liquid Al.	m.p. 660 >700	9 12
Al-steel	Formation of iron aluminides.	500	9
Co-TaC	Directionally solidified eutectic; dissolution and re-precipitation.	1 200	9
Cu-W	No interaction up to melting point.	1 083	9
Fe-W	Formation of Fe_7W_6: dissolution of fibre.	1 000	9
Mg(AZ91)-C	No significant reaction at m.p. of alloy provided O and N avoided during infiltration.		11, 14
Ni-Al_2O_3	Slight reaction to pit fibre. $NiAl_2O_4$ spinel forms in air.	1 100 1 100	9
Ni-C(I)	Ni activates recrystallisation of fibre.	1 150–1 300	9
Ni-(II)	Ni activates recrystallisation of fibre.	800	9
Ni-SiC	Formation of nickel silicides.	800	9
Ni-W	Recrystallisation of fibre. Degradation of creep properties.	1 000 900	9
Ti-B	Formation of TiB_2.	750	9
Ti-SiC	Formation of TiC, $TiSi_2$ and Ti_5Si_3.	700	9

laminates of layers of fibres and metal foil is also effective. Another successful method of fabricating continuous fibre composites is to pre-coat the individual fibres with metal-matrix by PVD prior to consolidation by hot pressing. Very fine composite microstructures can be made by co-deforming a dispersed metal in a metal matrix, for example, Cu-Cr, Cu-Nb, in these cases very high strengths can be achieved for large drawing strains.

The properties which can be achieved in composites include higher strength, higher stiffness, improved high-temperature properties, lower or matched coefficients of thermal expansion and improved wear resistance. This is usually at the expense of conventional ductility but in most cases improved fracture toughness can be obtained. The following tables illustrate what has been achieved in aluminium, magnesium, titanium, zinc and copper alloys. The improvement in strength is often larger in the weaker alloys with less benefit being realised in the stronger alloys, e.g. see Table 37.3. All the alloys show an improvement in elastic modulus when reinforced.

Table 37.3 MECHANICAL PROPERTIES OF ALUMINIUM ALLOY COMPOSITES AT ROOM TEMPERATURE

Base alloy	*Nominal composition*	*Form*	*Heat treatment*	*% particulate*	*0.2% proof stress* MPa	*Ultimate tensile stress* MPa	*Elongation* %	*Elastic modulus* GPa	*Fracture toughness* MPa $m^{-1/2}$	*Density* g cm^{-3}	*References*
6061	Mg 1.0 Si 0.6 Cu 0.2 Cr 0.25	Extrusion	T6	Nil	276	310	20.0	69.9	29.7	2.71	1, 2
				10% Al_2O_3	297	338	7.6	81.4	24.1	2.81	1, 2
				15% Al_2O_3	317	359	5.4	87.6	22	2.86	1, 2
				20% Al_2O_3	359	379	2.1	98.6	21.5	2.94	1, 2
				13% SiC	317	356	4.9	89.5	17.9	—	1
				20% SiC	440	585	4.0	120.0	—	—	3
				30% SiC	570	795	2.0	140.0	—	—	3
2014A	Cu 4.4 Mg 0.7 Si 0.8 Mn 0.75	Extrusion	T6	Nil	414	483	13.0	73.1	25.3	2.80	1, 2
				10% Al_2O_3	483	517	3.3	84.1	18.0	2.92	1, 2
				15% Al_2O_3	476	503	2.3	91.7	18.8	2.97	1, 2
				20% Al_2O_3	483	503	0.9	101.4	—	2.98	1, 2
				10% SiC	457	508	1.8	91.2	17.7	—	1
		Sheet	T6	8.2% SiC	448	516	4.5	82.5	—	—	1
2219	Cu 6.0 Mn 0.3 V 0.1	Extrusion	T6	Nil	290	414	10.0	73.1	—	—	1, 2
				15% Al_2O_3	359	428	3.8	88.3	—	—	1, 2
				20% Al_2O_3	359	421	3.1	91.7	—	—	1, 2
2618	Cu 2.0 Mg 1.5 Si 0.9 Fe 0.9 Ni 1.0	Sheet	T6	~ 10% SiC	396	468	3.3	93.6		—	1, 6
		Extrusion	T6	Nil	320	400	—	75.0	—	—	1, 6
		Extrusion	T6	13% SiC	333	450	6.0	89.0	28.9	—	1, 6
7075	Zn 5.6 Mg 2.2 Cu 1.5 Cr 0.2	Extrusion	T6	Nil	617	659	11.3	71.1	—	—	6
				12% SiC	597	646	2.6	92.2	—	—	6
8090	Li 2.5 Cu 1.3 Mg 0.95 Zr 0.1	Extrusion (18 mm)	T6	Nil	480	550	5.0	79.5	—	—	1, 6
			T6	12% SiC	486	529	2.6	100.1	—	—	1, 6

Table 37.4 MECHANICAL PROPERTIES OF ALUMINIUM ALLOY COMPOSITES AT ELEVATED TEMPERATURES

Base alloy	*Nominal composition*	*Form*	*Heat treatment*	*% particulate*	*Temperature* °C	0.2% *proof stress* MPa	*Tensile strength* MPa	*References*
6061	Mg 1.0 Si 0.6 Cu 0.2 Cr 0.25	Extrusion	T6	15% Al_2O_3	22	317	359	1
				15% Al_2O_3	93	290	331	1
				15% Al_2O_3	150	269	303	1
				15% Al_2O_3	204	241	262	1
				15% Al_2O_3	260	172	179	1
				15% Al_2O_3	316	110	117	1
				15% Al_2O_3	371	62	69	1
2014A	Cu 4.4 Mg 0.7 Si 0.8 Mn 0.75	Extrusion	T6	15% Al_2O_3	22	476 (413)	503 (483)	1
				15% Al_2O_3	93	455 (393)	490 (434)	1
				15% Al_2O_3	150	407 (352)	434 (379)	1
				15% Al_2O_3	204	317 (283)	338 (310)	1
				15% Al_2O_3	260	200 (159)	214 (172)	1
				15% Al_2O_3	316	103 (62)	110 (76)	1
				15% Al_2O_3	371	55 (35)	55 (41)	1

Figures in parentheses are for basic alloy without particulate.

Table 37.5 MECHANICAL PROPERTIES OF MAGNESIUM ALLOY COMPOSITES AT ROOM TEMPERATURE

Base alloy	*Nominal composition*	*Form*	*% reinforcement*	0.2% *proof stress* MPa	*Tensile strength* MPa	*Elongation* %	*Elastic modulus* GPa	*References*
ZK60A	Mg 5.5 Zn 0.5Zr	Extruded rod	Nil	260	325	15.0	44	5
			15% SiC (partic.)	330	420	4.7	68	5
			20% SiC (partic.)	370	455	3.9	74	5
			15% SiC (whisker)	450	570	2.0	83	5
			20% B_4C (partic.)	405	490	2.0	83	5
	MG 12Li	Squeeze infiltration	Nil	—	80	8.0	—	7
		Squeeze infiltration	12% Al_2O_3(fibre)	—	200	3.5	—	7
		Squeeze infiltration	24% Al_2O_3(fibre)	—	280	2.0	—	7
		Extruded rod	Nil	—	75	10.0	45	7
		Extruded rod	20% SiC(whisker)	—	338	0.8	112	7

Table 37.6 MECHANICAL PROPERTIES OF TITANIUM ALLOY COMPOSITES

Base alloy	*Form*	*% reinforceant*	*Temperature* °C	0.2% *proof stress* MPa	*Tensile strength* MPa	*Elongation* %	*Elastic modulus* GPa	*References*
Ti-6Al-4V	Particulate	10% TiC	21	800	806	1.13	106–120	4
	Particulate	10% TiC	427	476 (393)	524 (510)	1.70 (11.6)	—	4
	Particulate	10% TiC	538	414 (359)	455 (441)	2.40 (8.5)	—	4
	Particulate	10% TiC	649	369 (221)	317 (310)	2.90 (4.2)	—	4
	Particulate	10% B_4C	21	—	1 055 (890)	—	205	5
Ti-6Al-4V	SCS6 fibre	35% SiC	20	—	1 634 (1 024)	1.1	213	15
Ti-15Al-3V	SCS6 fibre	35% SiC	20	—	1 572 (772)	0.7	210	15

Figures in parentheses are for basic alloy without reinforcement.

Table 37.7 MECHANICAL PROPERTIES OF ZINC ALLOY COMPOSITES AT ROOM TEMPERATURE

Base alloy		*Nominal composition*	*Form*	*Tensile strength* MPa	*Elongation* %	*Elastic modulus* GPa	*References*
ZA-12	Zn-12% Al	As cast	Nil	300	5	87	8
		As rolled	Nil	350	22	115	8
			10% SiC	323	0	119	8
			20% SiC	373	0	129	8
		Extruded	Nil	313	—	102	8
			20% Al_2O_3	532	—	120	8
ZA-27	Zn-27% Al	As cast	Nil	410	2	73	8
			10% SiC	396	0	92	8
			20% SiC	330	0	110	8
			50% SiC	310	0	220	8
		As rolled	Nil	393	16	85	8
			10% SiC	370	3	89	8
			20% SiC	349	0	102	8
		Extruded	Nil	340	—	83	8
			12% SiC	382	—	105	8
			16% SiC	466	—	130	8
			Nil	314	—	78	8
			20% Al_2O_3	382	—	100	8

Table 37.8 MECHANICAL PROPERTIES OF CO-DEFORMED COPPER COMPOSITES AT ROOM TEMPERATURE

System	*Ultimate strength* MPa	*Drawing strain*	*References*
Cu-15wt% Cr	906	7.0	17
Cu-18vol% V	1 850	11.0	18
Cu-30vol% Fe	1 500	8.5	18
Cu-12vol% Nb	400	3.1	19
Cu-12vol% Nb	810	8.2	19
Cu-12vol% Nb	1 180	10.3	19
Cu-12vol% Nb	1 520	11.9	19
Cu-20vol% Nb	430	3.1	19
Cu-20vol% Nb	970	8.2	19
Cu-20vol% Nb	1 410	10.3	19
Cu-20vol% Nb	1 840	11.9	19

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